

Docker installation on Windows 11

To install Docker on Windows 11, download the **Docker Desktop** installer from the official website [Docker Get Started](#). Run the installer as an administrator and enable **WSL 2** (Windows Subsystem for Linux) when prompted. Finally, restart your PC and launch Docker Desktop to start using it.

Step 1: Check System Requirements

Ensure your PC meets the following minimum requirements:

- Windows 11 64-bit (Home, Pro, Enterprise, or Education).
- At least 4 GB of RAM.
- Virtualization enabled in your system's BIOS settings.

Step 2: Download Docker Desktop

1. Go to the Docker Get Started page.
2. Click the **Download for Windows** button to get the Docker Desktop Installer.exe file.

Step 3: Install Docker Desktop

1. Locate the downloaded .exe file in your Downloads folder, right-click it, and select **Run as administrator**.
2. When the configuration window appears, make sure the **"Use WSL 2 instead of Hyper-V"** checkbox is selected (this is the recommended option).
3. Follow the remaining on-screen prompts to complete the installation.
4. Once finished, click **Close and restart** to apply the system changes.

Step 4: Verify the Installation

1. After your PC restarts, open Docker Desktop from the Start menu or the desktop shortcut.
2. Accept the **Docker Subscription Service Agreement**.
3. Open a **Command Prompt** or **PowerShell** window and run the following command to verify your installation:

```
docker --version
```

Docker installation on Ubuntu

To install Docker Engine on Ubuntu, the recommended approach is to set up **Docker's official repository** to ensure you get the latest stable version.

Here is the complete step-by-step installation guide.

1. Clean Up Old Versions

Before starting, remove any conflicting or unofficial packages from your system:

```
for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove $pkg; done
```

Use code with caution.

2. Set Up the Repository

Install the necessary system prerequisites to allow apt to utilize packages over HTTPS:

```
sudo apt-get update
sudo apt-get install ca-certificates curl
```

Use code with caution.

Next, add Docker's official GPG key to verify package signatures:

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Use code with caution.

Add the stable repository to your system's apt sources configuration:

```
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Use code with caution.

3. Install Docker Packages

Update your local package index and install the core Docker components and plugins:

```
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

Use code with caution.

4. Verify the Installation

Run the official hello-world image to confirm Docker is working correctly:

```
sudo docker run hello-world
```

Use code with caution.

5. Run Docker Without Sudo (Optional)

To run Docker commands without sudo, add your user to the docker group:

```
sudo usermod -aG docker $USER
```

```
newgrp docker
```

```
docker run hello-world
```

Docker installation on Mac OS

To install Docker on macOS, download [Docker Desktop](#) for either Apple Silicon (M1/M2/M3) or Intel chips. Open the downloaded .dmg file and drag the Docker icon into your Applications folder. Launch the app, accept the license, and follow the on-screen prompts.

1. Download and Install

- Go to the official [Docker Get Started](#) page.
- Click on **Download Docker Desktop** and choose the correct chip architecture for your Mac (Apple Silicon or Intel Chip).
- Wait for the .dmg file to download (usually over 600MB).
- Open the downloaded file and drag the **Docker** icon into the **Applications** folder.

2. Launch Docker Desktop

- Open your Applications folder or use Spotlight to launch **Docker.app**.
- The macOS system will ask for confirmation to open a downloaded application; click **Open**.
- Accept the Docker Subscription Service Agreement.
- When prompted to complete the installation, choose **Use recommended settings**.
- Enter your Mac's administrator password to authorize the installation of privileged components.

3. Verify the Installation

- Open your Terminal and run the following commands to ensure everything is working correctly:

```
docker --version
```

Use code with caution.

- Test your setup by running the sample "hello-world" container:

```
docker run hello-world
```